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MSc THESIS

MicroRNA target prediction with novel machine learning methods beyond quadratic attention

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Επιστήμη Δεδομένων και Τεχνολογίες Πληροφοριών

ΔΙΠΛΩΜΑΤΙΚΗ ΕΡΓΑΣΙΑ

Πρόβλεψη στόχων μικροRNA με νέες μεθόδους μηχανικής μάθησης πέρα από τις τετραγωνικές αρχιτεκτονικές

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ABSTRACT

MicroRNAs are small, non-coding RNA molecules that partake in a variety of regulatory mechanisms, with the most well-studied being their role in the suppresion of translation of mRNAs via binding to sequences called MicroRNA Response Elements (MREs). They are thus implicated in a wide range of mechanisms in physiology and their disregulation can be decisive for disease states. In the current age of multi-omics analyses and personalized medicine, miRNAs are discovered via high-throughput sequencing experiments such as small RNA-seq and their regulatory roles can be studied via experiments such as AGO-eCLIP and AGO-CLASH that rely on the quantification of miRNA-mRNA hybrids. Given that a miRNA can target several different mRNA molecules, it is often complex to discern the regulatory effect of many microRNAs from a single high-throughput sequencing experiment which necessitates the use of advanced computational analyses for this purpose. To that end, several machine learning and deep learning models have emerged with the goal of miRNA target prediction. Such models aim either to predict valid miRNA-MRE pairs, or make predictions at the gene level, either finding possible MREs within the entire gene/mRNA sequence or using this information to predict changes in gene expression with the integration data from additional high-throughput sequencing experiments regarding the transriptome. The goal of this thesis is to develop a machine/deep-learning model that, given pairs of microRNAs and MREs predicts which ones would interact. For this purpose, AGO2-eCLIP data were utilized for training and evaluation of several models leading to the development of an weighted ensemble of machine learning models that is able to discern valid from invalid such pairs with an Average Precision score comparable to other state of the art models trained on the same task. The output of the model also allows for advanced interpretability via providing scores for the contribution of each feature to the final prediction.

SUBJECT AREA: Thesis LaTeX template

KEYWORDS: list, of, keywords

ΠΕΡΙΛΗΨΗ

Τα microRNAs είναι μικρά, μη κωδικοποιούντα μόρια RNA που συμμετέχουν σε διάφορους ρυθμιστικούς μηχανισμούς. Σε αυτή την εργασία, αναπτύσσουμε μοντέλα μηχανικής μάθησης για την πρόβλεψη αλληλεπιδράσεων μεταξύ microRNAs και στόχων mRNA. Χρησιμοποιώντας δεδομένα AGO2-eCLIP, δημιουργήσαμε ένα weighted ensemble μοντέλο που επιτυγχάνει υψηλή απόδοση σε μια ισορροπημένη κατανομή των δεδομένων κατά οικογένεια microRNA.

ΘΕΜΑΤΙΚΗ ΠΕΡΙΟΧΗ: Πρότυπο διπλωματικής εργασίας LaTeX

ΛΕΞΕΙΣ ΚΛΕΙΔΙΑ: μικροRNA, πρόβλεψη στόχων, μηχανική μάθηση

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ACKNOWLEDGMENTS

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PREFACE

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1. INTRODUCTION

1.1 MicroRNAs

MicroRNAs are the most well characterized class of non-coding RNA molecules, most commonly between 18-22 nucleotides long, that act mainly to regulate the post-transcriptional landscape of the cell, functioning as small guide RNAs for proteins of the Argonaute (AGO) family. Typically, miRNAs bind to the target mRNA sequence via their seed region which covers positions 2-8 in their sequence. Perfect complementarity leads to AGO mediated cleavage of the target with the Argonaute proteins functioning as endonucleases. Consequently, the translation of the target mRNA molecule halts and it is further targeted by other cellular RNA nucleases that lead to its complete degradation. In that manner, they are able to function in a vast variety of regulatory networks that control cellular behaviour. MicroRNAs are named based on the miRBase scheme whereby a 3-letter code denotes the species the miRNA was discovered in (e.g hsa: Homo sapiens), followed by miR in the case of mature sequences (or mir for precursor ones) and finally a number that conveys when the molecule was discovered/published with respect to the other entries (AMBROS *et al.* 2003; Kozomara & Griffiths-Jones 2011). As a rule, miRNAs from different organisms that are orthologous share the same name after the species identifier, whereas paralogous sequences have a single-letter suffix that distinguishes them and hairpin loci that give rise to the same mature miRNA are specified by a single-digit suffix. Finally, in cases where two different molecules are excised from the same precursor, a 5p or 3p suffix is used. It is also worth noting that with the advent of advanced high-throughput sequencing techniques, it has become plausible to detect isoforms of miRNAs. Such molecules differ from the canonical miRNA sequence either in terms of length of their 5' and/or 3' ends or because their sequence has been modified via processes like A-to-I or C-to-U editing. These modifications subsequently change how they are loaded to AGO and how they interact with target MREs (Tomasello *et al.* 2021).

1.1.1 Biogenesis

The transcription of a miRNA gene is performed by RNA Polymerase II or RNA Polymerase III and leads to the production of a precursor called pri-miRNA which is longer in length than the mature molecule and has the structure of a stem-loop with single-stranded overhangs extending from its 3' and 5' ends. It is necessary that a stem of a pri-miRNA molecule is of specific size, its loop is at least ten base pairs long and its ends can produce single-stranded RNAs, so that it can be processed into a mature miRNA (Zeng & Cullen 2003; Zhang & Zeng 2010). The maturation process starts with a protein complex comprising of an endoribonuclease called Drosha and a protein called DGRC8. DGRC8 is a protein that binds double-stranded RNA, in this case the stem of the pri-miRNA, stabilizing its interaction with Drosha. The cleavage of the pri-miRNA by Drosha is also guided by DGRC8 which recognizes the region 11 base pairs downstream from its binding site.

The product is pre-miRNA, a molecule that is 60-70 base pairs long and that is exported from the nucleus by exportin 5. An alternative origin for miRNA genes is introns of protein coding genes, such genes are also called mirtrons. Due to the fact that the pri-miRNA molecules originate from introns,they do not have the appropriate structure needed in order to be processed by Drosha and DGRC8. A protein called DRB1 solves the lariat structure of intronic RNA and pre-miRNA is the immediate product of this process. Another example of Non-canonical miRNA biogenesis includes 7-methyl Guanosine (m7G)-capped pre-miRNA, which are exported to the cytoplasm via Exportin 1 without being cleaved by Drosha first. All pre-miRNAs are recognized by a type 3 RNAse called DICER and TRBP which is a protein that binds the stem of the pre-miRNA. DICER can then cleave the pre-miRNA molecule into two leading to products of 22 base pairs long, one of which will be loaded into the RISC complex that is comprised of DICER, TRBP, AGO, PACT, and TNRC6A-C (O'Brien *et al.* 2018).

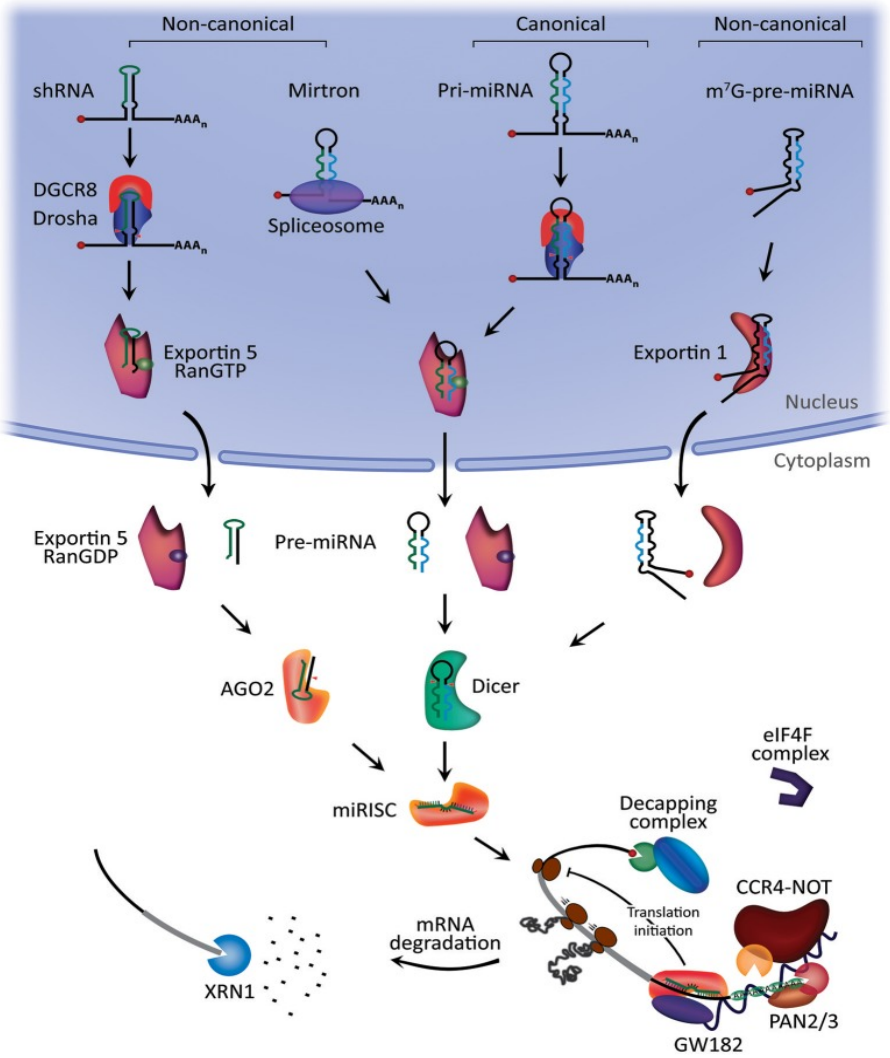

Figure 1.1: The steps of miRNA biogenesis. (O'Brien *et al.* 2018)

1.1.2 Target Binding

The strand that is loaded is to be used as a guide RNA for RISC, via hybridizing with the mRNA. The dsRNA is mainly formed between the seed region of the miRNA (nucleotides 1-8) and the 3'UTR of the mRNA, with additional, downstream base pairing aiding the interaction. Given perfect complementarity, the endonuclease activity of AGO2 is induced and the mRNA is cleaved. At the same time, proteins of the GW182 complex are recruited in order to act as scaffolds for necessary effector proteins such as PAN2-PAN3 and CCR4-NOT complexes that function as Poly-A Deadenylases that remove the protective 3' polyA tail, given the interaction between tryptophan from W repeats of the GW182 complex and the PABC complex on the mRNA has been established. Finally, removal of the 5'cap of the mRNA from DCP2 takes place followed by 5'-3' degradation from the exonuclease XRN1.

Most of the miRNA:MRE complexes in the cell are not perfectly complementary and additional 3' base pairing plays an important role for their stabilization. The regulatory efficiency scales with the number of seed matches according to the following order (Grimson *et al.* 2007):

1. 8mer: 8 matches within the seed region (1-8)
2. 7mer m8: 7 matches within the seed region (2-8)
3. 7mer A1: 7 matches within the seed region with the first nucleotide being an Adenine (1-7)
4. 6mer: 6 matches within the seed region (2-7)
5. 6mer offset: 6 matches within the seed region (3-8)
6. 5mer: 5 matches within the seed region (2-6)

As mentioned above, additional base pairing can enhance the stability of the complex and is classified as supplementary binding if it does so without being necessary for the regulatory role of the miRNA, or as compensatory if it compensates for mismatches within the seed region. Finally, there is a seedless mode of pairing for miRNAs that has been described in C.elegans which likely occurs due to AU rich sequences in the seed region. Regulatory efficiency is also affected by relationships between different binding sites and binding events within the same mRNA molecule. Adjacent sites for different miRNAs and repetitive sites for the same miRNA can both function in a cooperative manner. Functional cooperativity refers to the event whereby many different binding events have a cumulative effect, whereas binding cooperativity to the physical interaction of different RISC complexes supporting each other in the binding of their target (Diener *et al.* 2024).

1.1.3 Additional mechanisms of action

Binding of miRNAs on MREs can also increase the efficiency of translation instead of suppress it. A protein called FXR1 is necessary for this phenomenon, which in combination with RISC may bind specific sequence motifs (such as AU Rich Elements, AREs) or other sequences in the 5'UTR leading to activation of translation. MicroRNAs can also regulate gene expression via utilizing other, albeit less studied, mechanisms. For example, RISC complexes can be transported in the nucleus in order to exert regulation on transcription. In order for this to occur, they have to be imported from the cytoplasm via importin 8, importin α/β or XPO1. Within the nucleus, these complexes can bind to DNA regulatory elements affecting the binding of transcription factors and of other regulatory proteins (such as chromatin remodelling complexes) on them, thus having an impact on the transcription of related genes (Kim *et al.* 2008; Chaluvally-Raghavan *et al.* 2016; Xiao *et al.* 2017; Guo *et al.* 2021). A noteworthy example is that of miR-30b-5p which upon starvation, translocates to the nucleus in order to bind regulatory elements known as CLEAR elements that are known to bind the TFEB transcription factor. In that manner, it suppresses the transcription of these genes and inhibits lysosomal biogenesis and the autophagic flux (Guo *et al.* 2021). Additionally, RISC has been shown to be able to bind to mRNAs within the nucleus and induce their degradation while AGO and Drosha also participate in the splicing of premature mRNAs (O'Brien *et al.* 2018).

1.1.4 Roles in Physiology and Disease

Given their diversity regarding both sequence and mechanisms of action, miRNAs partake in many physiological and pathological states. Due to their capability of regulating multiple genes in a variety of different manners, they are important components in developmental processes that require the fine-tuning of gene expression at precise time intervals, with a prominent example being that of miR-196. Anterioposterior polarity is mediated by the SHH protein in forelimb and hindlimb buds and its expression requires Retinoic Acid (RA) signalling induced Hoxb8 expression. Hoxb8 is a transcription factor protein that is upregulated as an immediate-early response to ectopic RA administration to forelimb but not to hindlimb buds due to a specific miRNA, miR-196, that inhibits the translation of Hoxb8 in a tissue specific manner (Hornstein *et al.* 2005). MicroRNAs are also able to regulate protein synthesis spatiotemporally within the cell, localizing to specific subcellular compartments before exerting their effects (Corbin *et al.* 2009).

Deregulation of miRNA expression can lead to or deteriorate a big number of disease states ranging from congenital defects to autoimmune diseases, neurological disorders and many types of cancer. For instance, Adam *et al.* 2009 showed that miR-200 family expression in UMUC3 cells lead to increase in E-Cadherin expression and elevated sensitivity to EGFR inhibitors. MiRNAs can also be excreted from one cell to be taken in by a target via exosomes. Lin *et al.* 2019 demonstrated that miR-21 is excreted from bladder cancer cells via exosomes that are taken in by macrophages in order to activate the PI3K/AKT pathway and promote cancer progression by polarizing them towards the M2

phenotype. In addition, miRNAs can be detected in fluid samples from patients and can thus be used as biomarkers for specific diseases. For instance, Sapre *et al.* 2016 showed that a combination of 6 miRNAs from urine are able to predict Urothelial Carcinoma of the Bladder (UCB) patients with a high AUC score (0.85). Finally, miRNAs can be used as targets for therapeutics that either induce or reduce their expression. An example is miR-222 which has elevated expression levels in cardiomyocytes after specific types of exercise as well as following exercise rehabilitation after heart failure. This elevated expression leads to targeting of p27, HIPK-1, HIPK-2, and HMBOX1 which promotes cardiomyocyte hypertrophy, proliferation, and survival (Ding *et al.* 2017).

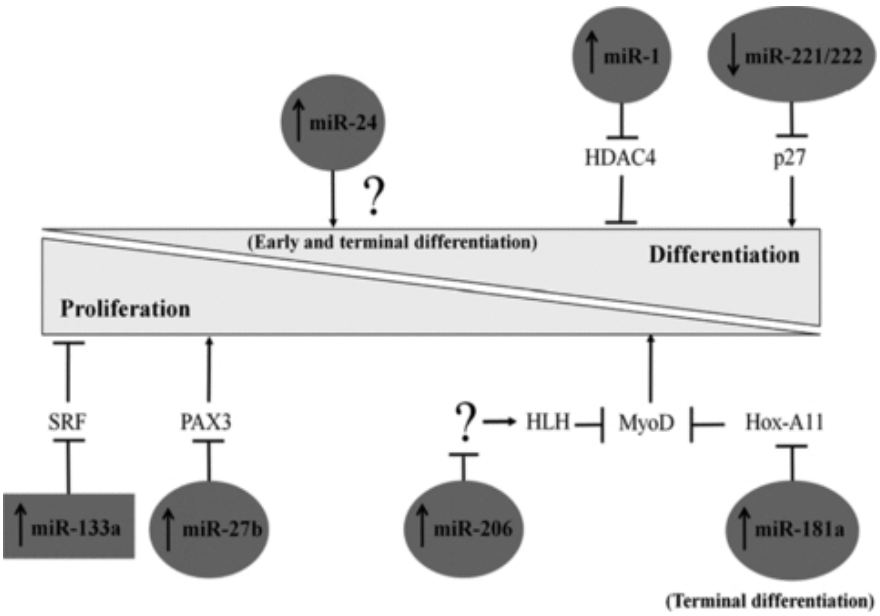

Figure 1.2: MiRNAs regulating muscle differentiation. (Güller & Russell 2010)

1.1.5 High-throughput sequencing methods for miRNA research

In the era of multiomics studies, it has become prevalent to study the totality of the miRNA part of the transcriptome as well as their targets via the use of high-throughput sequencing techniques. Such techniques allow for the detection and quantification of all the miRNA molecules in a tissue as well as of the complexes that these form with MREs. The most widely used one is small RNA-seq which includes isolation of RNA, cDNA library construction and sequencing. During RNA isolation, the total RNA might be extracted from a tissue, optionally followed by enrichment methods such as length separation or binding of small RNAs to to specific proteins. Library construction involves a reverse transcription step followed by ligation of adapters and/or polyadenylation (Benesova *et al.* 2021). The following image depicts the double ligation small RNA-seq protocol:

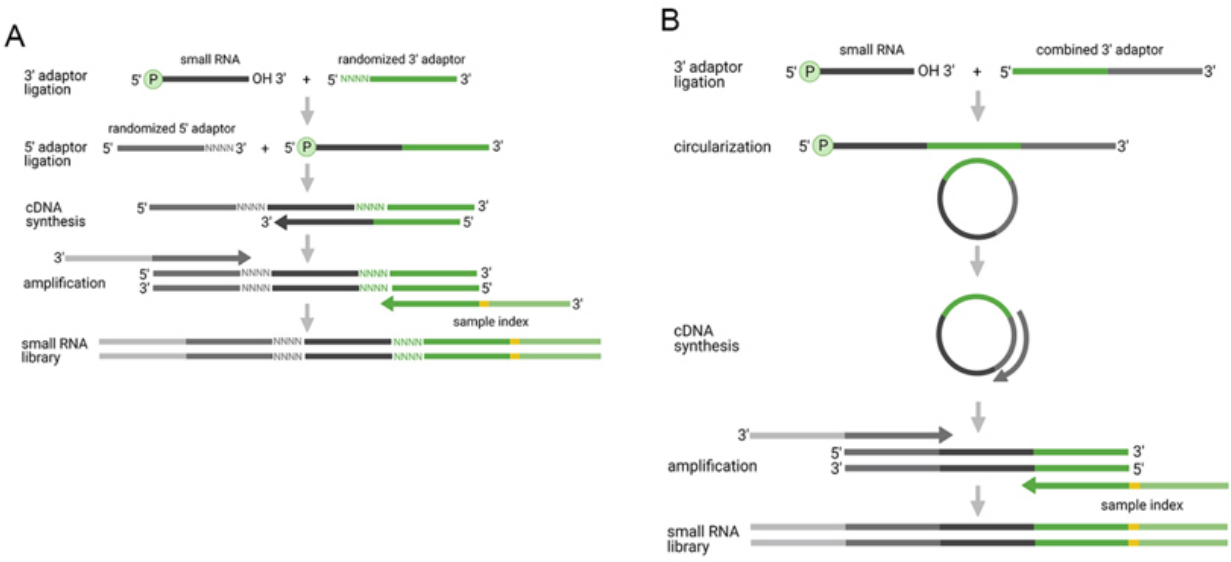

Figure 1.3: Double ligation small RNA-seq. (Benesova *et al.* 2021)

Types of experiments that capture and quantify miRNA:MRE complexes are used to study the binding sites of miRNAs and include AGO-eCLIP and AGO-CLASH. The latter involves the following steps (A *et al.* 2013):

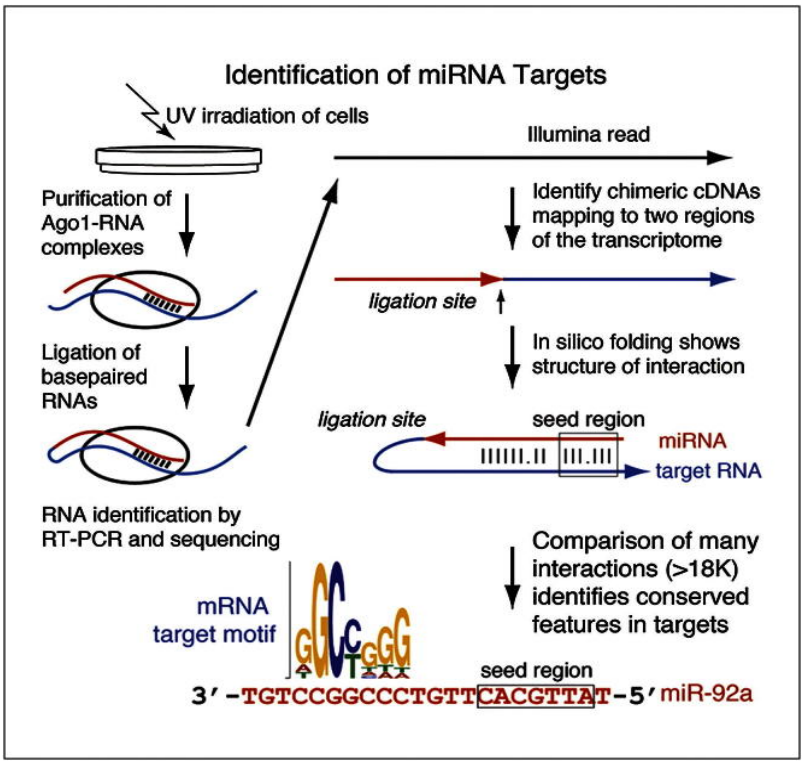

Figure 1.4: AGO-CLASH. (A *et al.* 2013)

Before cross-linking via UV irradiation, AGO expression is induced via the use of Doxicyclin, where the concentration is monitored in order for AGO to be close to endogenous levels. Despite that, differences between endogenous expression do occur and with them, the risk of bias regarding specific interactions. After cross-linking, hydrolysis of interacting RNA molecules takes place followed by ligation of the hybridized miRNA:MRE duplex which creates hybrid molecules to be reverse transcribed, amplified and sequenced. AGO-eCLIP follows a different procedure that does not require modifications to the endogenous expression. During AGO-eCLIP, the RISC:MRE complex is cross-linked using UV irradiation and the RNA is fragmented in the same manner as with AGO-CLASH, then immunoprecipitation, using antibodies that bind the AGO protein, follows and ultimately ligation of the miRNA 5'-P to the mRNA's 3'-OH group along with adapter ligation to the mRNA's 5' end (Manakov *et al.* 2022). Due to the immunoprecipitation step, no exogenous induction of expression is needed, eliminating the aforementioned bias and providing an image that is more physiologically relevant.

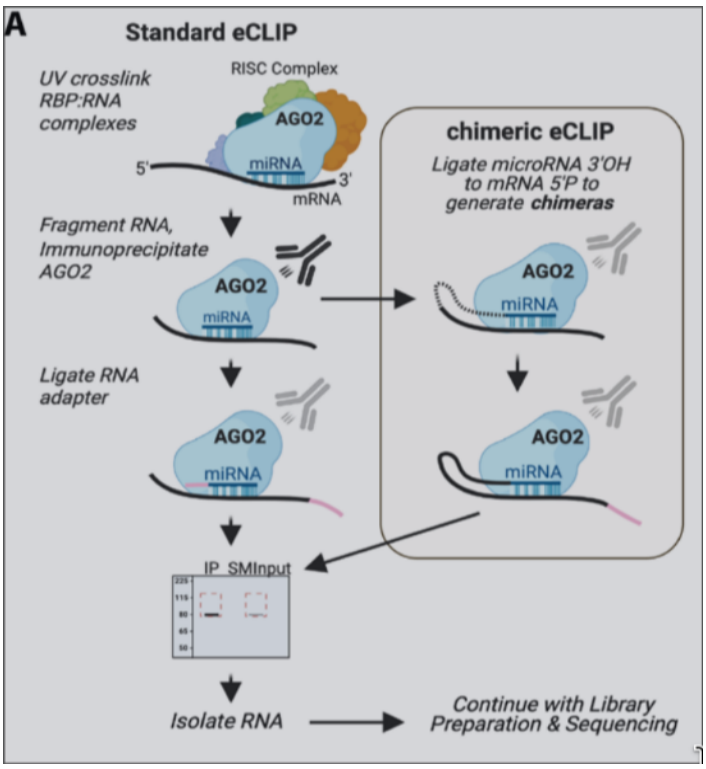

Figure 1.5: AGO-eCLIP. (Manakov *et al.* 2022)

There is also a family of techniques used to quantify the regulatory effect of miRNAs on their targets, which includes Degradome Sequencing, Parallel Analysis of RNA Ends and Genome-Wide Mapping of Uncapped and Cleaved Transcripts (Gregory *et al.* 2008; Addo-Quaye *et al.* 2008; German *et al.* 2009). In these experiments, the polyA transcripts are isolated from the tissue of interest, then ligation of the first adapter occurs only on transcripts with 5' monophosphate (which are the ones that have lost their 5'm7G cap due to miRNA mediated decay), followed by reverse transcription, 3' adapter ligation, PCR and

sequencing. The result is depicted as a peak on an mRNA transcript where the x-axis shows the sequence the miRNA targeted and the y-axis the frequency of 5' ends found at this position.

High-throughput sequencing techniques produce enormous amounts of data regardless of their end goal. Commonly, the immediate result from the sequencing instrument involves files that are called FASTQ files which include entries of all the sequenced molecules along with a quality score that conveys the reliability of the base that was sequenced at a given position. Each entry of a FASTQ file follows the format below:

1. 1st Line: @ character followed by an identifier for the entry
2. 2nd Line: Raw nucleotide seq
3. 3rd Line: + character optionally followed by the sequence identifier again
4. 4th Line: quality values for the sequence

It is worth noting that there is a number of different sequencing instruments that can be used for the same protocol but utilize different biochemistry principles in order to perform sequencing at such large scales. Due to these differences, the end result will be different even if one were to sequence the exact same samples and as such specific methods have been developed to mitigate this issue and ensure reproducibility across studies. Such methods are necessary when one wants to combine results from a number of different studies that have performed these experiments and the process called Batch Effect Correction. Finally, even within the same experiment it is heavily suggested that there are more than one replicates for the same sample in order to account for technical variation in the measurements.

Depending on the sequencing protocol and the end goal, these files are processed following different pipelines which include the usage of a series of computational tools. A common step regardless of the protocol is the Quality Control step (QC) in which various plots are produced in order to examine the integrity of the sequenced molecules based on the quality scores themselves as well as knowledge regarding the expected nucleotide and length distributions. Depending on the results, the next step involves filtering based on specific thresholds and optional removal of the adapter sequences followed by another QC step to ensure integrity has been maintained.

The next step is to map the sequences in the filtered file on the human genome/transcriptome and it is the first source of heterogeneity pertaining to such pipelines. Tools that perform this mapping are called Aligners and there is a big number of such tools each of which excels in specific circumstances. For small RNA-seq, two aligners that are commonly used are Bowtie2 and BWA-MEM (Langmead *et al.* 2009; Li 2013), that achieve high speed alignment of short reads on genomic sequences. The choice of the alignment tool as well as that of its alignment parameters are vital for the quality of the end result which necessitates their inclusion in the pipeline for reproducibility. A vital aspect of this step is the genome version that is used where the reads are to be aligned, as different versions lead

to differences in results. Most studies use hg38 which is considered the more up to date and complete version, although with deeper and more accurate sequencing more recent versions like the Telomere to Telomere version have emerged which might come to replace hg38, especially for the analysis of repetitive regions such as centromeres, telomeres and ribosomal array sequences. For AGO-eCLIP and AGO-CLASH, a typical aligner would not suffice as the reads that are sequenced are chimeric molecules, consisting partly of miRNA and of mRNA fragments. Thus, specific tools have been developed that can appropriately align each part, with an example being Hyb (Travis *et al.* 2014). Hyb begins by aligning the reads either to the genome or to a known database of molecules (ideally mature mRNAs) in order to identify reads that are non-contiguously aligned and then utilizes folding algorithms to annotate the recovered duplexes.The end result is a list of RNA:RNA hybrids.

1.2 miRNA Target Prediction Algorithms

Several methods have been developed over the years that utilize the data produced by the aforementioned experiments in order to achieve several goals related to elucidating miRNA biology. In general, predictive algorithms are separated in terms of the type of algorithms they use in order to make the prediction as well as the prediction task itself.

- Algorithms
  - Heuristic
  - Physics Based
  - Machine/Deep Learning
- Task
  - Site Level
  - Gene/Functional Level
  - Pathway Level

1.2.1 Site Level predictors

In terms of algorithms, a tool is characterized as Heuristic if it makes predictions using preset rules that are based on already discovered interactions (Lewis, Shih, *et al.* 2003; Lewis, Burge, *et al.* 2005). The first release of TargetScan was heuristics based, as it searched for perfect seed complementarity in the 3'UTR of each target provided, extended it to include as many matches as possible and then used software like RNAFold to check for additional complementarity in the miRNA 3'UTR with the next 35 nucleotide of the mRNA (Hofacker *et al.* 1994) and assigned a Z score that is the basis for the final predictions.

Such tools are considered the simplest form of prediction tools but function as a good baseline for comparison with more complex models.

Physics based tools may disregard the biology behind the task and focus on expressing the problem as an energy minimization one when it comes to MRE prediction although more recent ones aim to strike a balance between using some heuristics for narrowing down some predictions in combination with energy minimization. A fine example of this is IntaRNA (Busch *et al.* 2008; Mann *et al.* 2017; Gelhausen *et al.* 2019; Wright *et al.* 2014; Miladi *et al.* 2019; Raden *et al.* 2020) which is a rather advanced tool that provides a lot of different modes and was used as part of the feature extraction in this thesis. IntaRNA in its simplest mode can be used to scan genomes for target sites of specific small RNAs using a heuristics based approach that can be modified via a different number of command line arguments. The next mode in terms of simplicity is called IntaRNA exact and it aims to minimize the following sum:

$$E_{hybrid}(I) + ED1(I) + ED2(I) \tag{1.1}$$

In this model, E_{hybrid} denotes the hybridization energy of the base pairs forming between the two distinct molecules. Meanwhile, E_D represents the accessibility energy, the energy required to break existing internal (intra-molecular) bonds so the target sequences can interact. The model evaluates specific inter-molecular binding patterns, such as helical stackings, bulges, and interior loops. Because internal base pairings are not strictly mapped out, IntaRNA's predictions are flexible enough to account for any potential structural arrangement during single-site interactions. The ensemble mode of IntaRNA is based on a partition function computation that aims to find the optimal target site among many on the target sequence such that:

$$\arg\max Z(S) = \exp\left(-\frac{E_{D1}(S)}{RT}\right) \exp\left(-\frac{E_{D2}(S)}{RT}\right) \sum \exp\left(-\frac{E_{hybrid}(I)}{RT}\right) \tag{1.2}$$

The site-specific partition function, $Z(S)$, is calculated by multiplying the Boltzmann weights of the accessibility penalties (E_{D1} and E_{D2}) by the sum of the Boltzmann weights for all hybridization interactions (I) at that site. Here, R represents the universal gas constant and T is the temperature (with $RT \approx 0.616$ at 37°C). This partition function is used to determine the ensemble energy of all interactions at a given site using the equation $E(S) = -RT \ln(Z(S))$. By calculating ensemble energy rather than relying on a single, static base-pairing model, this approach accounts for the inherent flexibility and dynamic nature of the interacting regions. Furthermore, summing the partition functions across all potential sites yields the global partition function, $Z_{all} = \sum Z(S)$. This allows for the calculation of the probability that a specific interaction I will form, defined as $P_E(I) = \exp(-E(I)/RT)/Z_{all}$. IntaRNA provides additional options that function like other widely used software for the same task like RNADuplex, TargetScan and RNAup (Lorenz, Bernhart, *et al.* 2011; Hofacker *et al.* 1994; Lorenz, Hofacker, *et al.* 2016; Turner & Mathews 2010).

1.2.2 Functional and Pathway Level predictors

The advancement of machine and deep learning algorithms has paved the way for the development of models that circumvent the need for feature engineering and go beyond site level prediction. Heuristic and Physical approaches as those described above, as well as classical machine learning algorithms, need to take as input a number of features among which they try to calculate weights and correlations so as to create a function that separates the samples into predifined categories or fit a regression line on a multi-dimensional distribution. This requires prior processing of the samples so as to extract such meaningful features utilizing domain knowledge for the specific task. Modern deep learning models do not require such intensive preprocessing as they are developed in order to process the nucleotide sequence directly and find more abstract patterns that correlate with the desired output. The first and simplest type of architecture that has been used is Convolutional Neural Networks (CNNs). CNNs allow for the processing of the nucleotide sequence outside of the tabular limitations of classical Machine Learning by reading it in predifined windows along which mathematical operations are performed with the goal of creating a feature map whereby pooling is performed in order to reduce overfitting and make the model translation invariant. The final part usually consists of a classification layer that is used to output a probability for a specific class.

Some modern architectures like the one from Zacharopoulou *et al.* 2024 employ such architectures combining them with ones like Graph Neural Networks (GNNs) in order to extract a unified score across an entire gene and use it as a feature for a meta learner that is trained on expression data to predict functional binding which goes beyond site level predictions. The most recent and advanced models perform what is called Pathway Level prediction, by taking advantadge of site/gene level predictors and GNNs, they predict pathway level interactions between mRNAs-miRNAs-Competing Endogenous RNAs (ceRNAs), making them more relevant for identifying which of these pathways are crucial for disease states (Chiu *et al.* 2015; Jin *et al.* 2024). Finally, modern models may also rely on Transformer based architectures which are able to treat DNA/RNA as a language and learn more intricate pattern that are not necessarily in proximity in the input sequence. Examples of these include the vanilla Transformer, Titans and the Mamba architecture which is not transformer based but has shown promise in DNA related tasks (Behrouz *et al.* 2024; Gu & Dao 2024)

1.2.3 Transformers

Transformers were introduced in the landmark paper ”Attention is All You Need” (Vaswani *et al.* 2023). First, the input sequence is tokenized and mapped to dense vector embeddings. To inject information about the relative or absolute position of the tokens in the sequence, positional encodings, which are either learned or based on sinusoidal functions, are added to these embeddings. Through learned linear projections, the input is mapped into three distinct matrices: Queries (Q), Keys (K), and Values (V). The scaled dot-product attention mechanism, which is vital to the architecture’s success, is defined

by the following formula:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V \tag{1.3}$$

Rather than computing a single attention function, Transformers employ Multi-Head Attention, projecting the queries, keys, and values into multiple subspaces, performing the attention function in parallel, and concatenating the outputs. The standard Transformer encoder consists of a stack of identical layers. Each layer contains two primary sub-layers: a Multi-Head Attention mechanism and a position-wise Feed-Forward Neural Network. Every sub-layer is encompassed by a residual connection followed by layer normalization, formulated mathematically as LayerNorm$(x + \text{Sublayer}(x))$.

The Decoder follows a similar structural philosophy to the Encoder but introduces critical modifications to facilitate autoregressive sequence generation. Each identical decoder layer comprises three primary sub-layers: a Masked Multi-Head Self-Attention mechanism, an Encoder-Decoder Multi-Head Attention mechanism (often termed Cross-Attention), and a position-wise Feed-Forward Neural Network. As with the Encoder, residual connections and layer normalization encompass each of these sub-layers.

The masking in the first sub-layer is an essential modification; it mathematically prevents sequence positions from attending to subsequent positions. By setting the scaled dot-product attention scores of future tokens to $-\infty$ prior to the softmax operation, the model ensures that the prediction for position i depends exclusively on the known outputs at positions strictly prior to i.

In the subsequent Encoder-Decoder Attention sub-layer, the queries (Q) are derived from the output of the previous decoder layer, while the keys (K) and values (V) are sourced directly from the final output of the Encoder. This cross-attention mechanism allows every position in the decoder to attend over the entire input sequence, effectively grounding the generated output in the original provided context. Finally, the output of the decoder stack is passed through a linear transformation and a softmax function to produce a probability distribution over the vocabulary for the next predicted token.

1.2.4 Titans

While Transformers have achieved unprecedented success across various domains, their standard architecture suffers from two fundamental limitations: the quadratic computational complexity of the attention mechanism with respect to sequence length ($\mathcal{O}(N^2)$), which severely bounds the maximum context window, and the static nature of their weights during inference. Because a trained Transformer's weights are frozen, it cannot learn from new context dynamically; it merely caches tokens in its finite working memory.

To address these bottlenecks, the Titans architecture (Behrouz *et al.* 2024) departs from relying exclusively on self-attention by introducing a dynamic, biologically inspired Neural Memory System. Rather than forcing a single attention mechanism to process the en-

tirety of a sequence, the Titans architecture partitions the cognitive workload across three distinct memory modules:

Short-Term Memory This module operates identically to the traditional Transformer attention mechanism. It serves as the model's immediate mental workspace, providing high-resolution, precise processing strictly for the local, immediate context.

Long-Term Memory This is the core innovation of the architecture. Instead of attempting to scale the context window to infinity, Titans introduce a secondary neural network whose weights are dynamically updated *during inference* (test time). This module compresses historical context into neural weights rather than caching explicit token representations.

Persistent Memory This tier consists of learnable, static parameters that encode the fundamental rules and meta-knowledge of the task itself, ensuring the model does not suffer from catastrophic forgetting while its Long-Term Memory aggressively adapts to novel prompts.

Mathematical Formulation of the Neural Long-Term Memory

The foundational mathematics of the Titans architecture redefines long-term memory not as a static cache of tokens, but as an online optimization problem. The independent neural network ($\mathcal{M}$) governing this memory adapts its parameters at test time via gradient descent, driven by an associative loss function and a biologically inspired "surprise" metric.

The long-term memory module aims to learn associations between historical keys (k_t) and values (v_t) generated from the input sequence. The memory network $\mathcal{M}$, parameterized by weights at time step $t-1$ (W_{t-1}), attempts to reconstruct the value given the key: $\mathcal{M}(k_t; W_{t-1}) \approx v_t$. The accuracy of this reconstruction is measured by an associative loss function, typically defined as Mean Squared Error (MSE):

$$\mathcal{L}_t = \|\mathcal{M}(k_t; W_{t-1}) - v_t\|^2 \tag{1.4}$$

Drawing from cognitive science, Titans posit that highly predictable data should require little memorization, whereas unexpected data requires structural changes in the neural pathways. Mathematically, "surprise" is quantified as the gradient of the associative loss with respect to the memory parameters. If the memory module flawlessly predicts the value (low loss), the gradient is near zero (low surprise). Conversely, a failed prediction yields a large gradient (high surprise):

$$S_t = \nabla_{W_{t-1}} \mathcal{L}_t \tag{1.5}$$

Instead of remaining frozen after training, the memory module updates its parameters sequentially during inference. To prevent memory overflow and catastrophic forgetting,

the update mechanism incorporates a decay factor (α) for past gradients and a learning rate (θ) for the momentary surprise (S_t). This creates a momentum-based update rule:

$$W_t = W_{t-1} - (\alpha \Delta W_{t-1} + \theta S_t) \tag{1.6}$$

where ΔW_{t-1} represents the accumulated historical surprise. By embedding this test-time gradient descent directly into the forward pass, the Titans architecture effectively compresses the entirety of the sequence history into the deep weights of W_t. When a new query (q_t) arrives, the model simply queries this dynamically updated network ($\mathcal{M}(q_t; W_t)$) to retrieve context, entirely circumventing the $\mathcal{O}(N^2)$ dot-product computations required by standard attention mechanisms.

1.2.5 Mamba

While the Titans architecture addresses the context bottleneck by compressing history into dynamic weights, an alternative and highly successful paradigm has emerged in the form of Structured State Space Sequence Models (SSMs), most notably the Mamba architecture (Gu & Dao 2024). Mamba completely discards the attention mechanism, offering a fundamental architectural shift that scales linearly in time ($\mathcal{O}(N)$) with respect to sequence length, while maintaining a constant memory footprint during autoregressive generation.

Mathematical Foundation of Structured State Space Models

At its core, Mamba is inspired by continuous-time control theory. It maps a 1-dimensional input sequence $x(t) \in \mathbb{R}$ to an output $y(t) \in \mathbb{R}$ through an implicit latent state $h(t) \in \mathbb{R}^N$.

The classical continuous-time SSM is defined by a system of linear ordinary differential equations:

$$h'(t) = Ah(t) + Bx(t) \tag{1.7}$$

$$y(t) = Ch(t) \tag{1.8}$$

To process discrete tokens (such as text), this continuous system must be discretized. Using a timescale parameter Δ, the continuous matrices A and B are transformed into their discrete counterparts $\bar{A}$ and $\bar{B}$ (typically via the zero-order hold method). This yields a highly efficient recurrent formulation:

$$h_t = \bar{A}h_{t-1} + \bar{B}x_t, \quad y_t = Ch_t \tag{1.9}$$

Previous iterations of SSMs (such as S4) suffered from a fatal constraint: they were Linear Time-Invariant (LTI). Because their transition matrices (B, C, and Δ) were static and fixed across all time steps, they could not selectively focus on or ignore specific tokens, making tasks like targeted information retrieval or copy-pasting nearly impossible.

Mamba resolves this by introducing a *Selection Mechanism*. It parameterizes B, C, and the step size Δ to be input-dependent functions of the current token x_t:

$$B_t = \text{Linear}_B(x_t), \quad C_t = \text{Linear}_C(x_t), \quad \Delta_t = \text{softplus}(\text{Linear}_\Delta(x_t)) \tag{1.10}$$

By allowing Δ_t to vary based on the input, the model dynamically dictates the degree to which it should integrate the current token into the hidden state (h_t) or completely ignore it. This effectively grants the recurrent model the data-filtering capabilities of an attention mechanism, without the $\mathcal{O}(N^2)$ computational penalty.

Because Mamba's parameters are time-varying, the model cannot be computed using efficient Fast Fourier Transforms (FFTs) like its LTI predecessors. To overcome this, Mamba relies on a hardware-aware parallel scan algorithm. By fusing the scan operations and materializing the expanded hidden states (h_t) exclusively in the faster, on-chip SRAM of the GPU (bypassing the slower High Bandwidth Memory (HBM) IO bottleneck) Mamba achieves training speeds competitive with highly optimized Transformers while preserving the inference efficiency of a recurrent neural network.

1.2.6 Main Goal

The main goal of this thesis is to explore the use of architectures and algorithms that aim to surpass vanilla Transformers in terms of either performance or explainability. Thus, the Titans and Mamba architectures were employed and their results were compared to those of a vanilla Transformer deep neural network. Additionally, simpler machine learning models and their ensembles were employed in order to further examine the tradeoff between their higher degree of explainability and (potentially) speed and their performance on the Manakov *et al.* 2022 dataset.

2. FIGURES AND TABLES

2.1 Inserting tables

Below is a sample table with its caption. Make sure you put the caption **over** the table. For instance, this code in LaTeX:

```
\begin{table}
    % this is before the table
    \caption{A sample event log, sorted by timestamp}
    % this is to label and include it in our list of tables
    \label{tbl:event_log_sample}
    \begin{center} % center the table
    \begin{tabular}{c|c|c}
        \textbf{Case Id} & \textbf{Activity} & \textbf{Timestamp} \\
        \hline
        1 & Activity A & 2021-05-24T12:30:00Z \\
        1 & Activity B & 2021-05-24T12:30:26Z \\
        1 & Activity C & 2021-05-24T12:31:04Z \\
        2 & Activity A & 2021-05-24T13:19:39Z \\
        2 & Activity B & 2021-05-24T13:19:50Z \\
        3 & Activity A & 2021-05-24T13:19:59Z \\
        2 & Activity C & 2021-05-24T13:22:14Z \\
        3 & Activity B & 2021-05-24T13:24:24Z \\
        4 & Activity A & 2021-05-24T13:25:06Z \\
        3 & Activity C & 2021-05-24T13:25:27Z \\
        4 & Activity B & 2021-05-24T13:25:53Z \\
        4 & Activity C & 2021-05-24T13:27:48Z \\
        \hline
    \end{tabular}
    \end{center}
\end{table}
```

renders table 2.1:

Table 2.1: A sample event log, sorted by timestamp

Case Id	Activity	Timestamp
1	Activity A	2021-05-24T12:30:00Z
1	Activity B	2021-05-24T12:30:26Z
1	Activity C	2021-05-24T12:31:04Z
2	Activity A	2021-05-24T13:19:39Z
2	Activity B	2021-05-24T13:19:50Z
3	Activity A	2021-05-24T13:19:59Z
2	Activity C	2021-05-24T13:22:14Z
3	Activity B	2021-05-24T13:24:24Z
4	Activity A	2021-05-24T13:25:06Z
3	Activity C	2021-05-24T13:25:27Z
4	Activity B	2021-05-24T13:25:53Z
4	Activity C	2021-05-24T13:27:48Z

2.2 Inserting figures

Below is an example of inserting a figure with its caption. Make sure you put the caption **under** the figure. For example, this LaTeX code:

ABBREVIATIONS - ACRONYMS

AI	Artificial Intelligence
SPARQL	SPARQL Protocol and RDF Query Language
OWL	Web Ontology Language
OGC	Open Geospatial Consortium

APPENDIX A. FIRST APPENDIX

APPENDIX B. SECOND APPENDIX

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